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Sustainable **A**dvancement in **K**eeping **T**rash **O**rderly **(SAKTO);**

Cleaning Solution for Sustainable Waste Management

Summary

The Sustainable Advancement in Keeping Trash Orderly (SAKTO) project introduces an innovative, automated waste segregation system to improve waste management efficiency. By integrating smart technology, renewable energy, and sustainable materials, SAKTO aims to promote proper waste disposal and recycling while reducing environmental pollution.

SAKTO features an automated sorting system using infrared, inductive, and moisture sensors to classify waste into wet, dry, and metal categories. It incorporates solar power for energy efficiency, a touchless mechanism for hygiene, and real-time monitoring for optimized waste collection

This initiative supports the United Nations Sustainable Development Goals (SDGs) by addressing waste mismanagement, reducing landfill contributions, and promoting circular economy practices. Beneficiaries include government agencies, businesses, environmental organizations, waste management workers, educational institutions, and local communities.

By combining automation, sustainability, and community engagement, SAKTO provides a scalable, cost-effective, and environmentally responsible solution to modern waste management challenges, contributing to cleaner and more sustainable urban and rural environments.

Background of the Problem

Improper waste disposal poses significant environmental, health, and economic challenges globally. The World Bank (2018) projects that annual waste generation will reach 3.4 billion tons by 2050, with a substantial portion mismanaged, leading to pollution and biodiversity loss.

In the Philippines, daily solid waste generation is approximately 61,000 metric tons, with up to 24% comprising plastic waste. Inadequate collection systems and improper disposal result in significant pollution, particularly in marine environments. Despite the Ecological Solid Waste Management Act (RA 9003), enforcement and compliance remain weak, necessitating innovative waste management solutions.

Moreover, Davao Oriental, a province in the Davao Region, is facing a growing waste management crisis due to rapid urbanization, lack of proper collection systems, and improper disposal habits. Real-time monitoring data from DENR (2023) and local government sources show that municipalities like Governor Generoso, Cateel, and Baganga are particularly overwhelmed. These areas are battling rising waste volumes, clogged drainage, and public health concerns, making it crucial to adopt advanced solutions like SAKTO.

Despite efforts in waste management and environmental protection, current strategies remain insufficient in addressing proper waste segregation, disposal, and recycling. To combat the growing waste crisis, this project proposes SAKTO (Sustainable Advancement in Keeping Trash Orderly), an innovative solution designed to promote efficient waste disposal and real-time monitoring in sustainable waste management.

To address this issue, SAKTO is proposed as an innovative waste management solution. By integrating technology into waste disposal, SAKTO aims to promote proper waste segregation, incentivize responsible waste disposal, and reduce environmental pollution. By leveraging automated waste sorting, real-time data tracking, and community engagement, this initiative seeks to enhance local waste management efforts, ultimately contributing to a cleaner and more sustainable future.

Beneficiaries

The Sustainable Advancement in Keeping Trash Orderly (SAKTO) project introduces an innovative and efficient waste segregation system that enhances sustainability, hygiene, and waste management practices. This initiative aligns with key United Nations Sustainable Development Goals (SDGs), including SDG 3 (Good Health and Well-being) by reducing exposure to waste-borne diseases, SDG 11 (Sustainable Cities and Communities) by improving urban waste management, SDG 12 (Responsible Consumption and Production) by promoting efficient waste sorting and recycling, and SDG 13 (Climate Action) by minimizing landfill waste and its environmental impact. Below are its key beneficiaries:

Government and Policymakers. SAKTO provides governments with real-time data on waste management, supporting informed policy decisions and resource allocation. The system's efficiency improves overall waste management infrastructure, reducing the burden on local governments and enhancing public services. The data collected by SAKTO can be used to track progress towards sustainability goals and inform the development of effective

Figure 1. Government and Policymakers

waste management policies. This contributes to the achievement of national environmental targets and improves public health outcomes. SAKTO's success can serve as a model for replication in other municipalities, promoting broader adoption of sustainable waste management practices.

Figure 2. Environmental Organizations

Environmental Organizations. Environmental organizations can partner with SAKTO to promote sustainable waste management practices and monitor environmental impacts. The system's data-driven approach provides valuable information on waste generation and recycling rates, supporting evidence-based advocacy and policy recommendations. SAKTO's success serves as a case study for promoting sustainable solutions and inspiring similar initiatives. The reduction in pollution and improved environmental conditions align directly with the goals of environmental organizations. Partnerships with SAKTO enhance the effectiveness of environmental conservation efforts.

Figure 3. Businesses and Industries

Businesses and Industries. Businesses, particularly those generating significant waste, can leverage SAKTO to optimize waste management and reduce operational costs. The automated sorting system streamlines recycling processes, improving efficiency and potentially increasing the value of recyclable materials. SAKTO's data tracking capabilities provide valuable insights into waste generation patterns, informing waste reduction strategies and sustainable business practices. This can lead to cost savings through reduced waste disposal fees and improved resource management. By adopting SAKTO, businesses demonstrate their commitment to environmental responsibility, enhancing their brand image and attracting environmentally conscious consumers.

Figure 4. Waste Management Workers

Waste Management Workers. SAKTO significantly improves the working conditions of waste management workers by automating the physically demanding and often hazardous task of waste sorting. The touchless system reduces exposure to pathogens and hazardous materials, improving worker safety and health. The optimized waste collection schedules based on real-time fill-level monitoring reduces the workload and improves operational efficiency. This leads to a safer and more dignified work environment for waste management personnel. SAKTO's positive impact on worker well-being contributes to a more equitable and sustainable waste management system.

Local Communities. Local community benefit from SAKTO through improved environmental conditions, enhanced public health, and increased community engagement. The reduction in litter and improved hygiene contribute to a cleaner and more aesthetically pleasing environment. The system's convenience encourages community participation in waste management, fostering a sense of collective responsibility. SAKTO's success promotes social cohesion and strengthens community ties. Improved

Figure 5. Local Communities

environmental conditions and public health lead to a better quality of life for community members.

Figure 6. Educational Institutions.

Educational Institutions. Schools and universities can use SAKTO to educate students about sustainable waste management practices. The system's transparent operation demonstrates the importance of proper waste segregation and recycling. SAKTO's interactive features can be integrated into educational programs, fostering environmental awareness and promoting responsible citizenship. The hands-free operation promotes hygiene in educational settings, minimizing the spread of germs. By incorporating

SAKTO, educational institutions can lead by example, demonstrating a commitment to environmental sustainability.

Figure 7. Tourism Sector.

Tourism Sector. In tourist destinations, SAKTO's aesthetically pleasing design and efficient operation enhance the overall environment. The system's cleanliness and convenience contribute to a positive visitor experience. By demonstrating a commitment to sustainable waste management, tourist destinations can attract environmentally conscious travelers. Reduced litter and improved hygiene contribute to a more attractive and enjoyable environment for both residents and tourists. SAKTO's success can be showcased as a model of sustainable tourism practices, attracting investment and promoting economic development.

Households. Households directly benefit from SAKTO through improved waste segregation practices and enhanced hygiene. The touchless bin opening mechanism minimizes contact with waste, reducing the spread of germs and promoting a cleaner home environment. SAKTO's convenience encourages consistent and proper waste sorting, leading to reduced household waste and increased recycling rates. This contributes to a more

Figure 8. Households.

sustainable lifestyle at the individual level, aligning with responsible consumption and production principles. The real-time fill-level monitoring also optimizes waste collection, preventing overflowing bins and unpleasant odors.

SAKTO exemplifies a transformative approach to waste management by addressing the needs of policymakers, industries, communities, and environmental advocates. Through automation, sustainability, and data-driven waste segregation, it enhances operational efficiency, reduces pollution, and improves overall public health. By aligning with SDG 3, SDG 11, SDG 12, and SDG 13, SAKTO contributes to a cleaner, healthier, and more sustainable future while fostering innovation in waste management solutions.

The Proposed Solution of The Problem

The SAKTO (Sustainable Advancement in Keeping Trash Orderly) project introduces a smart, solar-powered waste segregation system that automates the sorting of wet, dry, and metal waste. Conventional waste disposal often results in inefficient recycling and increased landfill waste due to improper segregation. Manual sorting is time-consuming, unhygienic, and impractical, leading to environmental concerns. To address these challenges, SAKTO integrates Arduino-controlled automation, renewable energy, and sensor-based technology to create a more efficient and sustainable waste management system.

At the core of SAKTO's innovation is its automated waste identification system, which uses infrared (IR) and inductive sensors for accurate classification. Moisture sensors detect wet waste, while an inductive sensor identifies metal objects, ensuring proper sorting. Once categorized, waste is directed into designated compartments, reducing human error and improving recycling efficiency.

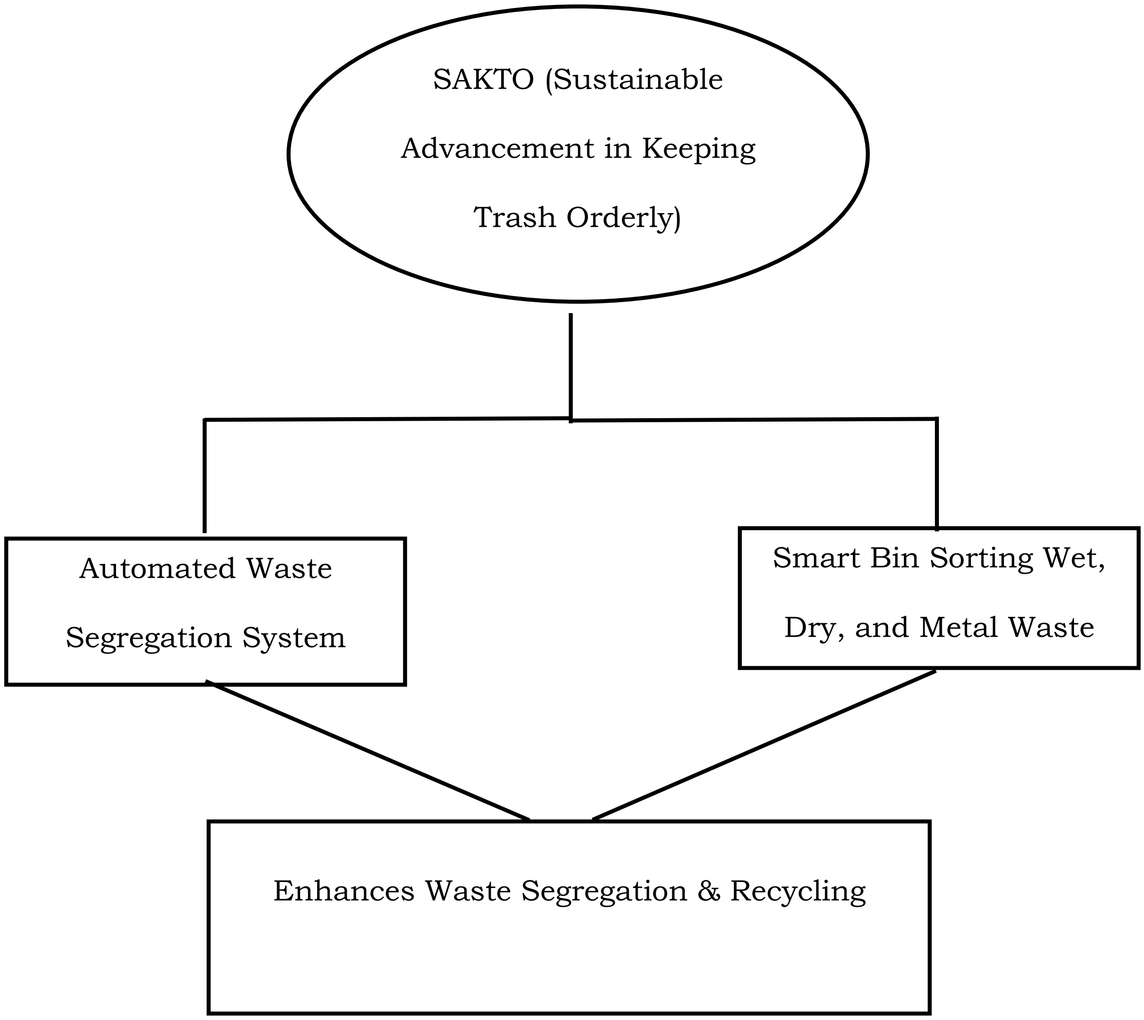
SAKTO also prioritizes hygiene and convenience by integrating an ultrasonic sensor-based touchless mechanism that automatically opens the bin lid when waste is detected. This hands-free system minimizes physical contact, reducing the risk of contamination, making it ideal for public spaces, schools, and households. An ultrasonic level sensor monitors each compartment's fill level, sending notifications when a bin is nearly full, optimizing collection schedules, and preventing overflows.

To enhance sustainability, SAKTO is solar-powered, utilizing renewable energy to minimize electricity consumption and make the system viable for off-grid locations. A rechargeable battery system ensures continuous operation even in low-light conditions. Additionally, SAKTO promotes eco-friendly construction by incorporating recycled materials into its design, aligning with environmental conservation efforts.

By integrating smart automation, renewable energy, and sustainable materials, SAKTO aligns with key Sustainable Development Goals (SDGs), including SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), SDG 13 (Climate Action), and SDG 15 (Life on Land). SAKTO provides an innovative, scalable solution to waste management challenges, fostering cleaner communities and a more responsible approach to waste disposal.

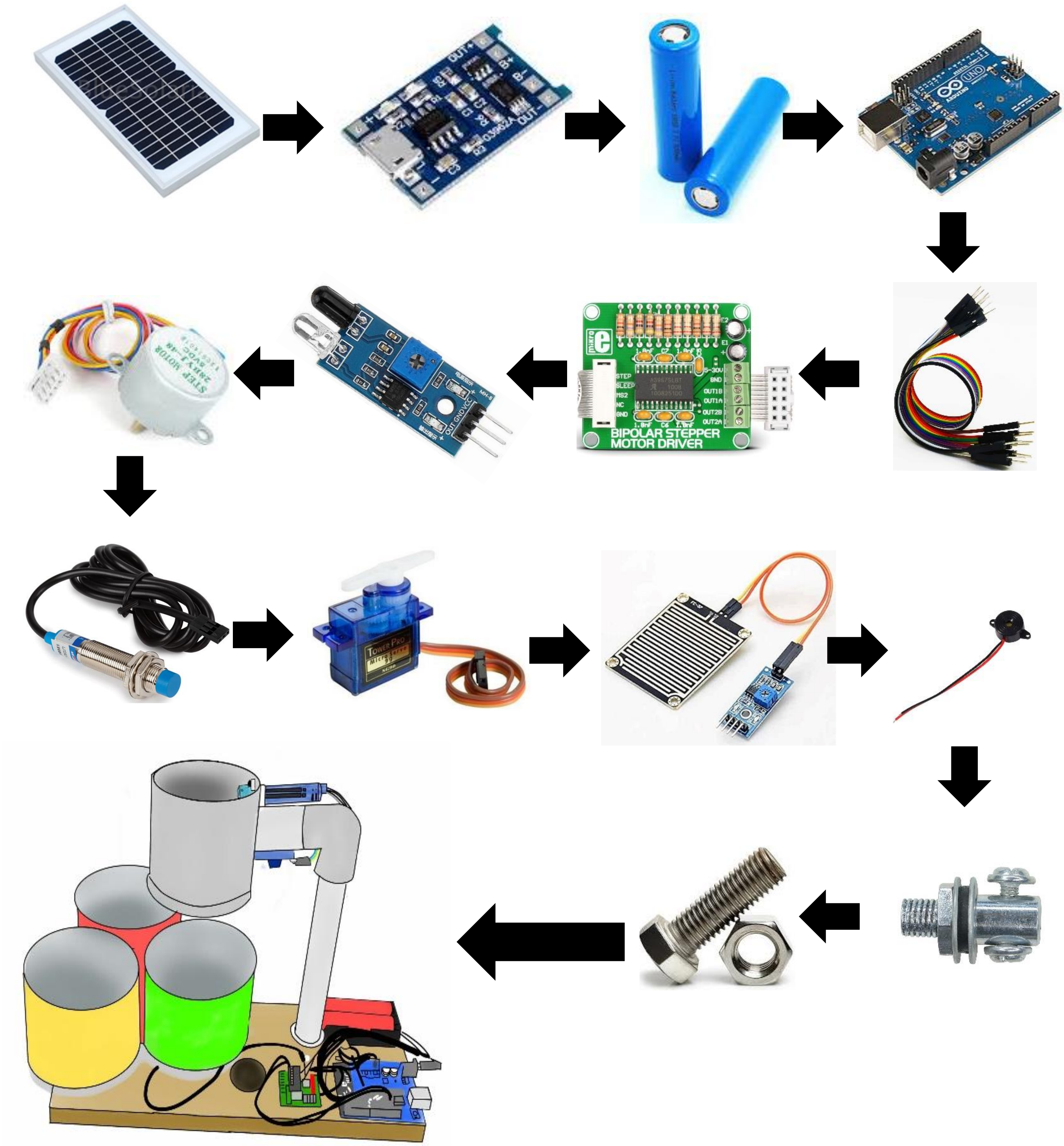
The conceptual framework, as shown in Figure 9, illustrates SAKTO's integration of automated waste segregation for efficient waste management. The Automated Waste Segregation System classifies waste into wet, dry, and metal categories using sensors, ensuring proper sorting and reducing landfill overflow.

Figure 9. The Conceptual Framework of the Proposed Solution



Methods / Details of The Proposed Solution

Figure 10. Flow Chart of SAKTO



The illustration in Figure 10 shows the process of how SAKTO is built, showing the materials that we will be utilizing for the functionality of the device. Providing this illustration, the following are the materials that are essential for SAKTO together with the discussion of its function in the process:

Figure 11. Solar Panel

Solar Panels

The solar panel converts sunlight into electrical energy, providing a renewable power source for the system. It ensures that SAKTO operates sustainably without relying on grid electricity, making it ideal for off-grid locations. By utilizing solar energy, the project reduces carbon emissions and promotes eco-friendly waste management.

Figure 12. TP4056

Figure 13. Li-ion Battery

Li-ion Battery

The Li-ion Battery serves as an energy storage unit, allowing the system to operate even when sunlight is unavailable. It provides continuous power to the Arduino and other electronic components, ensuring uninterrupted waste segregation functions. The rechargeable nature of the battery enhances the project's sustainability.

Figure 14. Arduino UNO

Arduino UNO

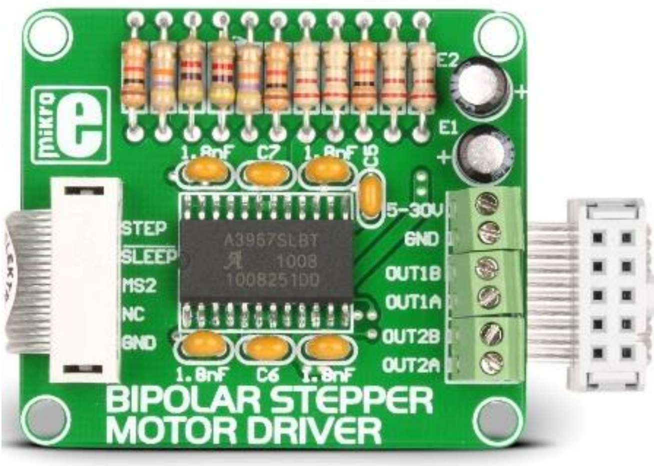
The Arduino UNO acts as the central processing unit, controlling all system operations, including waste detection, sorting mechanisms, and automation. It processes input from sensors and sends commands to motors and other components, enabling precise and intelligent waste management.

Figure 15. Jumper Wires

Jumper Wires

Jumper wires establish the necessary electrical connections between components, allowing data and power to flow seamlessly. They facilitate communication between sensors, motors, and the Arduino, ensuring coordinated operation. These wires are essential for assembling and maintaining the circuit.

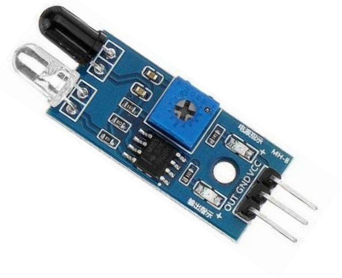
Figure 16. Stepper Motor Drive



Stepper Motor Drive

The stepper motor drive controls the movement of the stepper motors, regulating their speed and rotation. It ensures smooth and precise motor operation, which is crucial for the sorting mechanism's accuracy. By enabling controlled movements, it enhances the efficiency of waste segregation.

Figure 17. Infrared (IR) Sensors



Infrared (IR) Sensors

The Infrared (IR) Sensors detect waste materials based on their reflective properties, enabling the system to classify items. They play a key role in differentiating between wet, dry, and metal waste, improving sorting accuracy and reducing human error in waste management.

Figure 18. Stepper Motor

Stepper Motor

Stepper motor provides the mechanical movement needed for waste sorting. It rotates with high precision to direct waste into designated compartments, ensuring proper segregation. Its ability to move in controlled steps makes it ideal for automation.

Figure 19. Proximity Switch

Proximity Switch

The Proximity Switch senses objects near the bin and triggers automatic functions, such as opening the lid or activating the sorting system. This hands-free operation improves hygiene by reducing physical contact with waste bins, minimizing the spread of germs.

Figure 20. Servo Motors

Servo Motors

Servo motors control small but critical mechanical movements, such as adjusting flaps for waste redirection. Their quick response and precision contribute to the efficiency of the waste segregation system.

Figure 21. Rain Drop Sensor

Rain Drop Sensor

The Rain Drop Sensor detects moisture and alerts the system to environmental conditions. This helps protect electronic components from water damage and ensures the bin functions properly even in outdoor settings.

Figure 22. Buzzer

Buzzer

Buzzer provides audio notifications to users, signaling important alerts such as when the bin is full or when a sorting process is in progress. This feature enhances user awareness and promotes proper waste disposal habits.

Figure 23. Shaft Adaptor

Shaft Adaptor

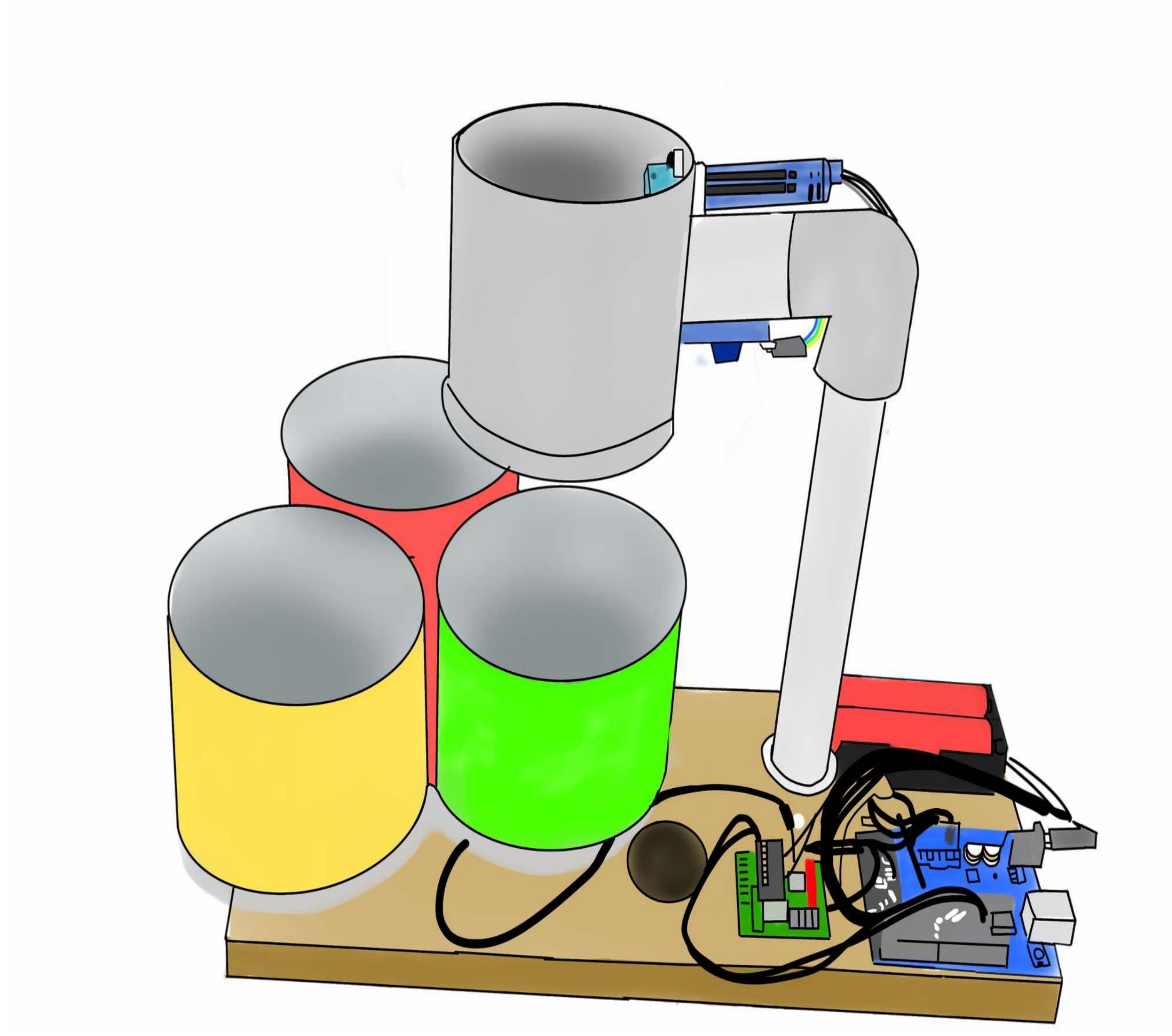
The shaft adaptor connects the stepper motor to the mechanical components, ensuring smooth and efficient movement. It enables a secure transfer of motion, allowing the sorting mechanism to function effectively.

Figure 24. Bolt and Nut

Bolt and Nut

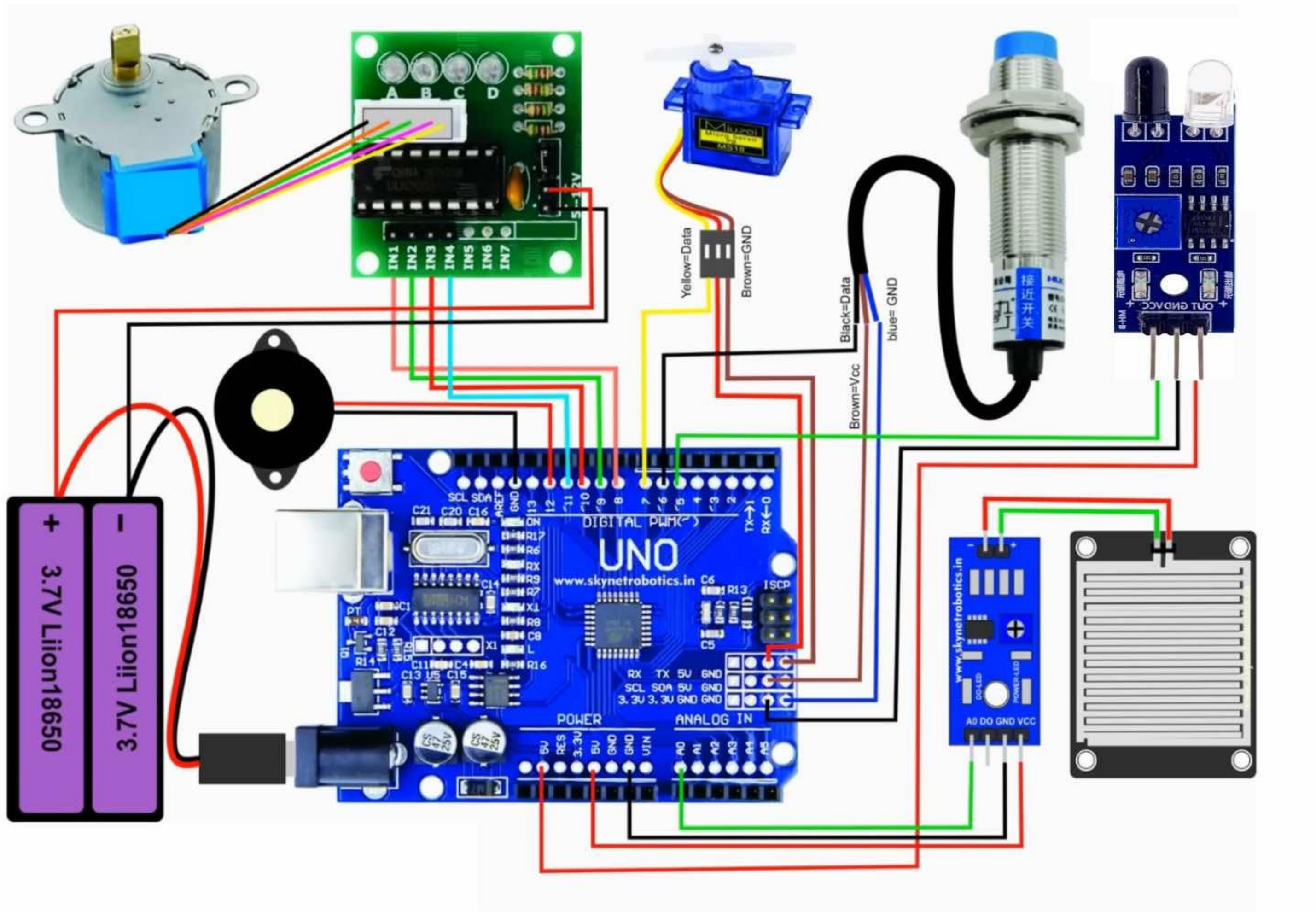
The Bolt and Nut secure the structural components of the SAKTO system, ensuring durability and stability. These mechanical fasteners hold critical parts together, preventing misalignment or malfunction during operation.

Figure 25. SAKTO External Prototupe



The illustration of SAKTO's external parts highlights the structural design and visible components that contribute to its functionality and user interaction. These external elements ensure durability, accessibility, and ease of use while complementing the automated waste segregation system. The outer framework houses the internal components, providing protection and structural support, while input and output mechanisms facilitate waste disposal and system notifications.

Figure 26. SAKTO Internal Prototype



The illustration of SAKTO's internal parts showcases the interconnected components that enable its automated waste segregation functionality. Each internal part plays a crucial role in ensuring the system's efficiency, accuracy, and sustainability. By integrating various sensors, motors, and a control system, SAKTO effectively sorts wet, dry, and metal waste while minimizing human intervention.

Features

SAKTO is an automated waste segregation system designed to enhance waste management efficiency while promoting sustainability. It integrates renewable energy, smart sensors, and a recycling initiative to optimize waste disposal and reduce environmental impact. Below are its key features:

1. Automated Waste Segregation System

SAKTO efficiently categorizes waste into wet, dry, and metal using advanced sensor-based detection and mechanical sorting mechanisms. Infrared (IR) sensors analyze surface properties to differentiate between dry and wet waste, ensuring precise categorization. Proximity sensors detect metallic objects by measuring conductivity and magnetic properties, allowing for accurate metal separation. Stepper motors and servo motors control sorting compartments, directing waste into designated bins. This automated process enhances efficiency and reduces the need for manual waste handling, promoting a cleaner and more organized waste management system. The integration of machine learning algorithms further refines the sorting accuracy by continuously improving waste classification based on collected data.

2. Solar-Powered, Off-Grid Operation

Designed for sustainability, SAKTO is powered by a solar panel, reducing dependence on grid electricity and making it ideal for off-grid areas, disaster-prone regions, and communities with limited power access. A TP4056 charge controller and Li-ion battery store energy, ensuring continuous operation even during low sunlight conditions. By utilizing renewable energy, SAKTO significantly lowers its carbon footprint, contributing to global efforts to combat climate change. This energy-efficient approach aligns with sustainable waste management practices and promotes environmental responsibility. Additionally, the system's modular design allows for easy maintenance and scalability, making it adaptable to various waste management needs in different communities.

3. Touchless Waste Disposal for Hygiene and Convenience

SAKTO incorporates a hands-free waste disposal mechanism to enhance sanitation and convenience. Motion sensors and ultrasonic sensors detect hand movements or waste placement, automatically opening the bin lid. This feature minimizes direct contact with waste, reducing the risk of bacteria and virus transmission. It is particularly beneficial for public spaces, healthcare facilities, schools, and commercial establishments where hygiene is a priority. By providing a safe and sanitary waste disposal method, SAKTO encourages proper waste management practices while protecting public health. The system also features an odor control mechanism, utilizing activated carbon filters to absorb unpleasant smells and maintain a fresh environment in waste disposal areas.

5. Sustainable and Eco-Conscious Construction

SAKTO is designed with sustainability at its core, utilizing recycled and locally sourced materials to minimize its environmental impact. By repurposing materials, the project not only reduces waste generation but also lowers manufacturing costs, making it an economically viable solution. This initiative aligns with Sustainable Development Goal (SDG) 12 – Responsible Consumption and Production, ensuring that resources are used efficiently while decreasing dependency on virgin materials. Through eco-conscious construction, SAKTO fosters a culture of sustainability in waste management infrastructure.

6. Public Awareness and Community Engagement

Beyond its technological capabilities, SAKTO serves as an educational tool to foster environmental awareness and promote responsible waste disposal habits. The project can be integrated with local government sustainability initiatives, allowing LGUs, schools, and public institutions to implement incentive-based waste management programs. Additionally, partnerships with environmental organizations and academic institutions can facilitate workshops, awareness campaigns, and training sessions on recycling, upcycling, and sustainability. By engaging communities and encouraging active participation, SAKTO aims to instill long-term behavioral change toward a cleaner and greener environment.

Processes

The development of SAKTO follows a structured approach, integrating automation, renewable energy, and smart waste segregation technologies. The process includes system design, material preparation, construction, electronics integration, programming, and deployment.

1. System Design and Planning

This phase ensures that SAKTO's design effectively integrates automated waste segregation with renewable energy and sensor-based sorting. The process begins by defining the waste segregation mechanism to accurately classify wet, dry, and metal waste. The electronic circuit schematic is then developed, incorporating solar power, battery management, sensors, motors, and microcontrollers to optimize performance. The smart bin's structural layout is carefully designed to determine sensor placement, motorized sorting, and power connections. Finally, the Arduino UNO is programmed to automate sorting, lid operation, and alert systems, ensuring seamless functionality and efficiency in waste management.

2. Material Sourcing and Component Preparation

To align with environmental conservation efforts, SAKTO prioritizes the use of recycled materials to reduce waste generation. The process begins with gathering essential electronic components, including IR sensors, inductive sensors, moisture sensors, servo motors, and Li-ion batteries. Structural materials, preferably recycled plastic or other sustainable alternatives, are sourced for the bin's casing to enhance durability and eco-friendliness. Before assembly, all sensors, motors, and electronic components undergo pre-testing to ensure functionality, reliability, and seamless integration into the system.

3. Frame and Bin Construction

Ensuring structural integrity and user accessibility, this phase focuses on building a durable and functional waste segregation system. The outer shell is constructed using recycled plastic, metal sheets, or wood to enhance sustainability and resilience. Inside, separate waste compartments for wet, dry, and metal waste are installed to maintain proper segregation. To improve hygiene and convenience, servo motor-controlled lids are integrated, enabling touchless waste disposal. Finally, internal wiring is securely fastened to protect it from environmental exposure, ensuring long-term system reliability. For added durability, the materials are treated with weather-resistant coatings, protecting the system from corrosion, moisture, and UV damage in various environmental conditions.

4. Sensor and Electronics Integration

This phase focuses on enhancing waste segregation accuracy and automation, reducing human intervention while improving efficiency. Infrared (IR) and inductive sensors are mounted to detect and classify waste, ensuring precise sorting of materials. Moisture sensors help distinguish between wet and dry waste, while a proximity sensor enables touchless lid operation for improved hygiene. An ultrasonic sensor monitors bin capacity and triggers alerts when the bin is full, optimizing waste collection schedules. Stepper and servo motors are secured to automate the sorting process, while all components are wired to the Arduino UNO and powered through the TP4056 charge controller, ensuring seamless operation.

5. Programming and System Testing

This phase focuses on intelligent automation, ensuring that SAKTO functions efficiently without manual intervention. The Arduino script is developed to process sensor inputs and control sorting mechanisms using if-else logic for accurate waste classification. A buzzer alert system is programmed to notify users when the bin reaches full capacity, optimizing waste management. Unit tests are conducted for individual components, followed by integrated system testing to identify and resolve errors, ensuring seamless operation and reliability in real-world conditions. To enhance performance, the system is also designed to update its firmware wirelessly, allowing for continuous improvements and the integration of new features without requiring physical access.

6. Solar Power System Setup

SAKTO's integration of solar power ensures energy efficiency and sustainability, allowing continuous operation even in off-grid locations. The solar panel is strategically mounted to maximize sunlight exposure, providing a renewable energy source for the system. A TP4056 charge controller regulates the charging process, preventing overcharging and ensuring battery safety. The Li-ion battery stores energy to power the sensors, motors, and

Arduino system, enabling uninterrupted functionality. Finally, the power supply is tested to ensure stable voltage delivery, optimizing system performance and longevity.

7. Final Assembly and Calibration

The final stage focuses on refining SAKTO's performance, durability, and user accessibility. All electronic components are secured in waterproof casings to protect against environmental elements, ensuring long-term functionality. Sensor sensitivity and motor speed are fine-tuned to improve sorting precision, reducing errors in waste classification. Automatic lid operation and bin compartmentalization are tested to ensure seamless functionality. To enhance user experience and promote proper waste disposal, clear labels are added to indicate wet, dry, and metal waste compartments, making the system intuitive and easy to use. Additionally, a mobile app integration is explored, allowing users to monitor bin capacity, receive maintenance alerts, and access waste disposal guidelines for a more interactive and efficient waste management experience.

8. Deployment and Monitoring

The final phase involves the real-world implementation of SAKTO to evaluate its effectiveness and optimize its performance. The system is installed in various environments, such as schools, public spaces, or community waste stations, to test its functionality under different conditions. User feedback and performance data are collected to assess sorting accuracy and overall efficiency. Based on real-world observations, sensor calibration and Arduino programming are adjusted to enhance precision. Performance improvements are documented, ensuring SAKTO continues to contribute to efficient waste management and sustainability while remaining adaptable for future upgrades and scalability.

By following these structured steps, the SAKTO project ensures the development of a smart, efficient, and eco-friendly waste segregation system. The integration of automated sorting, solar power, and sustainable materials contributes to a cleaner environment, enhanced waste management, and increased public awareness about proper waste disposal.

Code Integration for Arduino UNO

The Arduino UNO acts as the central controller in the SAKTO system, managing waste detection, classification, and sorting through integrated sensors and actuators. The microcontroller processes sensor inputs and controls servo motors, a buzzer, and other components to ensure efficient waste segregation.

Figure 27. Arduino code for SAKTO

```
#include <CheapStepper.h>
#include <Servo.h>
Servo servo1;
#define ir 5
#define proxi 6
#define buzzer 12
int potPin = A0; //input pin
int soil=0;
int fsoil;

CheapStepper stepper (8,9,10,11);

void setup()
{Serial.begin(9600);
  pinMode(proxi, INPUT_PULLUP);
  pinMode(ir, INPUT);
  pinMode(buzzer, OUTPUT);
  servo1.attach(7);
  stepper.setRpm(17);
  servo1.write(180);
delay(1000);
servo1.write(70);
delay(1000);

}

void loop()
{
  fsoil=0;
  int L =digitalRead(proxi);
Serial.print(L);
if(L==0)
{
  tone(buzzer, 1000, 1000);
 stepper.moveDegreesCW (240);
delay(1000);
servo1.write(180);
delay(1000);
servo1.write(70);
delay(1000);
stepper.moveDegreesCCW (240);
delay(1000);
}

if(digitalRead(ir)==0)
{
   tone(buzzer, 1000, 500);
   delay(1000);
  int soil=0;
  for(int i=0;i<3;i++)
    {
    soil = analogRead(potPin) ;
      soil = constrain(soil, 485, 1023);
        fsoil = (map(soil, 485, 1023, 100, 0))+fsoil;
          delay(75);
      }
```

```
  int L =digitalRead(proxi);
Serial.print(L);
if(L==0)
{
  tone(buzzer, 1000, 1000);
 stepper.moveDegreesCW (240);
delay(1000);
servo1.write(180);
delay(1000);
servo1.write(70);
delay(1000);
stepper.moveDegreesCCW (240);
delay(1000);
}

if(digitalRead(ir)==0)
{
   tone(buzzer, 1000, 500);
   delay(1000);
  int soil=0;
  for(int i=0;i<3;i++)
    {
    soil = analogRead(potPin) ;
      soil = constrain(soil, 485, 1023);
        fsoil = (map(soil, 485, 1023, 100, 0))+fsoil;
          delay(75);
      }
    fsoil=fsoil/3;
    Serial.print(fsoil);
    Serial.print("%");Serial.print("\n");

    if(fsoil>20)
           {
            stepper.moveDegreesCW (120);
            delay(1000);
              servo1.write(180);
              delay(1000);
              servo1.write(70);
                delay(1000);
              stepper.moveDegreesCCW (120);
                delay(1000);
              }

      else {
         tone(buzzer, 1000, 500);
           delay(1000);
            servo1.write(180);
              delay(1000);
              servo1.write(70);
                delay(1000);}
}

}

void loop()
{
  fsoil=0;
  int L =digitalRead(proxi);
Serial.print(L);
if(L==0)
```

Figure 28. Library Inclusion

```
#include <CheapStepper.h>
#include <Servo.h>
```

Library Inclusion

The Library Inclusion section includes the necessary libraries for the system's operation. The CheapStepper library enables control of the stepper motor, while the Servo library is used to manipulate the servo motor. These libraries provide predefined functions that simplify motor control, ensuring smooth and precise movements.

Figure 29. Variable & Pin Declarations

```
Servo servo1;
#define ir 5
#define proxi 6
#define buzzer 12
int potPin = A0; //input pin
int soil=0;
int fsoil;
```

Variable & Pin Declarations

In the Variable & Pin Declarations section, the code defines the input and output pins for various components, including the proximity sensor, infrared (IR) sensor, buzzer, and soil sensor. The potPin variable is assigned to the analog pin reading soil moisture levels, while soil and fsoil store sensor data. Using meaningful variable names improves readability and ease of modification.

Figure 30. Stepper Motor Initialization

```
CheapStepper stepper (8,9,10,11);
```

Stepper Motor Initialization

The Stepper Motor Initialization section creates an instance of the CheapStepper class, assigning specific pins (8, 9, 10, and 11) to control the stepper motor. This initialization ensures that the motor can be precisely rotated to facilitate mechanical waste sorting movements in the SAKTO system.

Figure 31. Setup Function

```
void setup()
{Serial.begin(9600);
  pinMode(proxi, INPUT_PULLUP);
  pinMode(ir, INPUT);
  pinMode(buzzer, OUTPUT);
  servo1.attach(7);
  stepper.setRpm(17);
  servo1.write(180);
delay(1000);
servo1.write(70);
delay(1000);

}
```

Setup Function

The Setup Function is executed once when the system starts. It initializes serial communication for debugging, configures pin modes for sensors and actuators, and attaches the servo motor to pin 7. Additionally, it sets the stepper motor speed to 17 RPM and initializes the servo position by moving it from 180° to 70°. These initial movements ensure that the system starts in a calibrated state, ready for operation.

Figure 32. Loop Function (Main Operations)

```
void loop()
{
```

Loop Function (Main Operations)

The Loop Function (Main Operations) continuously monitors sensor inputs and executes corresponding actions. It first resets the fsoil variable before checking the proximity sensor and IR sensor. If an object is detected, the system activates the buzzer and triggers motor movements. The loop ensures continuous real-time monitoring and automation of waste segregation.

Figure 33. Proximity Sensor Condition

```
  fsoil=0;
  int L =digitalRead(proxi);
Serial.print(L);
if(L==0)
{
```

Proximity Sensor Condition

In the Proximity Sensor Condition, the system reads the state of the proximity sensor. If an object is detected (L == 0), the buzzer sounds, and the stepper motor rotates 240° clockwise. The servo then moves between 70° and 180° before the stepper motor rotates back 240° counterclockwise. This movement sequence facilitates the processing of detected waste.

Figure 34. Infrared & Soil Sensor Condition

```
if(digitalRead(ir)==0)
{
```

Infrared & Soil Sensor Condition

The Infrared & Soil Sensor Condition monitors objects using an IR sensor. If an object is detected, the buzzer is activated, and the system proceeds to read the soil moisture sensor. This condition ensures that moisture-based waste segregation is performed only when necessary, optimizing system efficiency.

Figure 35. Soil Moisture Measurement & Mapping

```
int soil=0;
for(int i=0;i<3;i++)
  {
  soil = analogRead(potPin) ;
    soil = constrain(soil, 485, 1023);
      fsoil = (map(soil, 485, 1023, 100, 0))+fsoil;
        delay(75);
    }
    fsoil=fsoil/3;
    Serial.print(fsoil);
    Serial.print("%");Serial.print("\n");
```

Soil Moisture Measurement & Mapping

The Soil Moisture Measurement & Mapping section gathers soil moisture data by taking three consecutive readings from the sensor. The values are constrained within a range (485–1023) and mapped from 100% (high moisture) to 0% (dry). The final average value is displayed on the serial monitor, providing insight into the detected waste's moisture content.

Figure 36. Moisture-Based Actions

```
    if(fsoil>20)
          {
           stepper.moveDegreesCW (120);
           delay(1000);
             servo1.write(180);
             delay(1000);
             servo1.write(70);
               delay(1000);
             stepper.moveDegreesCCW (120);
               delay(1000);
             }

      else {
         tone(buzzer, 1000, 500);
           delay(1000);
            servo1.write(180);
              delay(1000);
              servo1.write(70);
                delay(1000);}
}

}
```

Moisture-Based Actions

The Moisture-Based Actions section dictates how the system responds based on soil moisture levels. If the detected moisture is above 20%, the stepper motor rotates 120° clockwise, the servo moves, and then the motor rotates back. If the moisture level is 20% or below, the buzzer sounds, and only the servo moves. These actions help categorize waste based on moisture content, assisting in efficient waste segregation.

Processes

The Arduino-based system for SAKTO automates waste segregation by utilizing a combination of sensors, motors, and actuators. This integration ensures an efficient and systematic process for sorting wet, dry, and metal waste. Below is a breakdown of how the code operates, highlighting its functionalities and interactions between components:

1. Sensor-Based Waste Detection and Classification

Sensors play a crucial role in SAKTO's waste detection and classification process, ensuring accurate and efficient sorting. The system employs multiple sensors, each serving a specific function to facilitate seamless automation. The Infrared (IR) sensor, connected to Pin 5, acts as the primary waste detection mechanism. It detects objects placed in the bin and triggers the sorting system. Once waste is identified, the system proceeds to classify its type for proper segregation.

To enhance accuracy, a proximity sensor, connected to Pin 6, ensures that the waste is correctly positioned before the sorting process begins. This prevents accidental activation of the system when no waste is present. Working in conjunction with the IR sensor, the proximity sensor validates waste presence and optimizes the sorting mechanism, reducing errors and improving overall efficiency.

2. Sorting Mechanism Control

Once waste is detected and classified, the sorting mechanism is activated to direct the waste into the appropriate compartment. The system relies on two key actuators: a stepper motor and a servo motor, each performing distinct roles in waste segregation.

The stepper motor, connected to Pins 8, 9, 10, and 11, controls the rotational movement of the waste sorting platform. It rotates either clockwise or counterclockwise, aligning the waste with the designated bin. For proximity-based sorting, the motor rotates 240°, while for moisture-based sorting, it adjusts by 120° to ensure precise segregation.

The servo motor, connected to Pin 7, is responsible for the opening and closing of waste bin lids. It moves to a predefined angle based on the detected waste type, ensuring proper disposal. Once the waste is sorted, the servo motor returns to its original position, resetting the system for the next waste input. This automated sorting mechanism enhances efficiency, minimizes human intervention, and improves waste management accuracy.

3. User Feedback and Notification System

The system provides real-time feedback to users through both auditory and visual indicators, ensuring efficient operation and user awareness.

The buzzer, connected to Pin 12, serves as an audible notifier. It emits a high-pitched tone when waste is detected, signaling users that the sorting process has started. Once sorting is completed, a short alert is sounded to confirm successful segregation. Additionally, if an error occurs—such as sensor failure or improper waste placement—the buzzer issues a distinct alert to notify the user of the issue.

The monitor output functions as a real-time display for system status and performance monitoring. It presents live data on sensor readings, waste classification, and sorting decisions, providing valuable insights into the system's operation. This output is particularly useful for debugging and optimizing performance, ensuring that SAKTO functions reliably in real-world applications.

4. Code Execution Process

The code governing SAKTO's automation is structured into two primary phases: system initialization and waste detection & sorting. This ensures seamless operation, proper sensor calibration, and efficient sorting mechanisms.

During system initialization (executed in the setup() function), all necessary input and output pins are configured. The proximity sensor, infrared (IR) sensor, buzzer, motors, and serial monitor are initialized to ensure smooth operation. The servo motor is positioned at its default state, ensuring that bin lids are properly closed before waste detection begins. Additionally, the stepper motor speed is set to 17 RPM, balancing efficiency and precision in waste sorting. Once the initialization is complete, the system enters standby mode, ready to detect and classify waste.

Waste Detection and Sorting (loop() function)

The loop() function is responsible for continuously monitoring waste input and executing sorting actions based on real-time sensor data. This ensures an automated, efficient, and accurate waste segregation process.

The process begins with initial waste detection, where the system continuously monitors the proximity and IR sensors. If waste is detected, the buzzer emits a short alert, signaling the start of the sorting sequence. The system then performs sorting based on proximity detection, where the stepper motor rotates 240° clockwise to position the waste at the sorting area, while the servo motor opens the corresponding bin for disposal. After the waste is sorted, the stepper motor resets by rotating 240° counterclockwise.

For sorting based on moisture content, sensor values are averaged, and the percentage is displayed on the serial monitor. If the moisture level exceeds 20%, the waste is classified as wet, prompting the stepper motor to rotate 120° clockwise and the servo motor to open the designated wet waste bin. If the moisture level is below 20%, the system follows the default sorting sequence for dry waste.

Finally, in the reset phase, the servo motor closes the bin lid, and the stepper motor returns to its starting position. The system then resets and enters standby mode, ready to detect and process the next waste item.

The integration of sensor-driven classification, motorized sorting, and real-time feedback mechanisms ensures precise segregation, reducing the risk of improper disposal. This systematic approach not only minimizes human effort but also contributes to a sustainable and environmentally responsible waste management system, fostering cleaner communities and promoting a circular economy.

Key Features of the Code

The SAKTO system integrates multiple technologies to create a highly efficient, self-operating waste segregation solution, combining automation, renewable energy, and smart connectivity to enhance waste management in various environments. Its key features include:

1. Automated Waste Detection and Sorting

The system utilizes infrared (IR) and proximity sensors to detect waste and determine its position before sorting. The IR sensor (Pin 5) detects when an object is placed in the bin, triggering the sorting mechanism. Meanwhile, the proximity sensor (Pin 6) verifies the correct positioning of the waste before proceeding, preventing accidental activation. This ensures an efficient and reliable classification process.

2. Stepper Motor Control for Waste Segregation

The stepper motor directs waste to the appropriate bin using precise rotations controlled by the Cheap Stepper library. It rotates 240° clockwise to move waste to the sorting area and an additional 120° CW if the waste is classified as wet. After sorting, it resets by rotating 240° counterclockwise, ensuring accurate and automated waste segregation.

3. Servo Motor Operation for Bin Opening

The servo motor (Pin 7) controls the opening and closing of bin lids. When the sorting process begins, the servo moves to 180° to open the bin lid, allowing waste to be disposed of. After the waste is placed, the servo returns to 70°, resetting the system for the next cycle. This controlled movement prevents unnecessary lid openings and ensures efficient sorting.

4. Moisture-Based Waste Classification

The moisture sensor, connected to Pin A0, determines whether waste is wet or dry by averaging three sensor readings and converting them into a percentage. If the moisture content exceeds 20%, the stepper motor rotates

120° CW to sort the waste into the wet bin; otherwise, it follows the default dry bin sequence. This feature enhances sorting accuracy and ensures effective waste classification.

5. User Feedback and Notifications

The system provides real-time user feedback through a buzzer (Pin 12) and a serial monitor output. The buzzer sounds for 1 second when waste is detected and emits a 0.5-second beep after sorting is completed, with an alert signal for sensor malfunctions. The serial monitor displays live data on waste type, sensor readings, and sorting decisions, allowing users to track performance and troubleshoot issues.

6. Optimized System Performance

The system operates in two key phases: initialization and continuous operation. The setup() function initializes sensors, motors, and communication interfaces while setting the servo motor to its default state. The loop() function continuously monitors waste input, executes sorting decisions in real time, and utilizes optimized delays to ensure smooth motor transitions, preventing overlaps and enhancing efficiency.

The SAKTO system provides an efficient automated waste segregation solution using sensor-based detection, motor-driven sorting, and user feedback mechanisms. By accurately identifying and classifying waste, it minimizes manual effort and promotes sustainable waste management. Its smart automation enhances efficiency, reliability, and environmental responsibility, contributing to a cleaner and more organized waste disposal system.

Figure 37. SAKTO Installed in a Public Area

SAKTO is strategically placed such as outside the mall, school, park, where people frequently dispose of their waste. The smart dustbin is powered by a solar panel and is equipped with sensors that detect different types of waste, ensuring proper segregation.

Figure 38. SAKTO Detects and Sorts Waste

When a person throws waste into the dustbin, the system’s sensors analyze the material. If the waste is wet, dry, or metal, SAKTO automatically directs it to the appropriate compartment.

After segregating the waste, SAKTO compacts certain materials for efficient storage. Once full, it sends a notification to the waste management team to schedule collection.

Table 1. Cost Analysis

ACTIVITIES	MATERIALS	UNIT	PRICE	QUANTITY	TOTAL
Body Structure	18650 Li-ion Batteries	pc	210	1	210
	Jumper wires	pc	98	1	98
	RainDrop Sensor	pc	74	1	74
	IR Sensor	pc	72	1	72
	Servo motor	pc	67	1	67
	Proximity switch	pc	94	1	94
	Stepper Motor	pc	86	1	86
	Stepper Motor Driver	pc	82	1	82
	Buzzer	pc	78	1	78
	Arduino uno	pc	218	1	218
Total					1079

The cost analysis for the SAKTO Smart project outlines the expenses required for its components, focusing on materials essential for its body structure and functionality. The total cost amounts to ₱1,079, covering various electronic and mechanical components necessary for automated waste segregation.

This cost breakdown highlights the affordability of SAKTO's design while maintaining efficiency in waste segregation. By integrating cost-effective materials with sustainable energy sources, the project ensures practicality and feasibility for community implementation.

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